CST1500 Computer Systems Architecture and Operating Systems

Workshop

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Lecture Videos and Slides

- Week Ten

VIDEOS

- Lecture 20 Protection
- Lecture 21 Network and Distributed Systems

SLIDES

- Lecture 20 Network and Distributed Systems
- Lecture 21 Introduction to Cloud Computing (A bit outdated)

CW1 Results – Re-submission

- If you received 19 or less marks, you did not pass the coursework.
- Good News! You can re-submit!
- Check the feedback, complete the work and submit by 13th December 2024 – Week 12.

- Use the following link in the Coursework One folder on the course page - Available now:

  Re-Submission - Coursework One - SOB Portfolio - Hendon Campus Re-Submission Link

- **Eligible Students ONLY can submit here!!**
- **If you did not submit anything, you cannot re-submit…**

Coursework 2 – Python Groupwork

- IMPORTANT
- If you have any issues with group members:
  - No response to messages.
  - Not getting work done on time.
  - Lack of engagement.
  - Any other issue.

- Let me or Ali know ASAP – in person or via email.

Attendance with Vevox

- Go To:

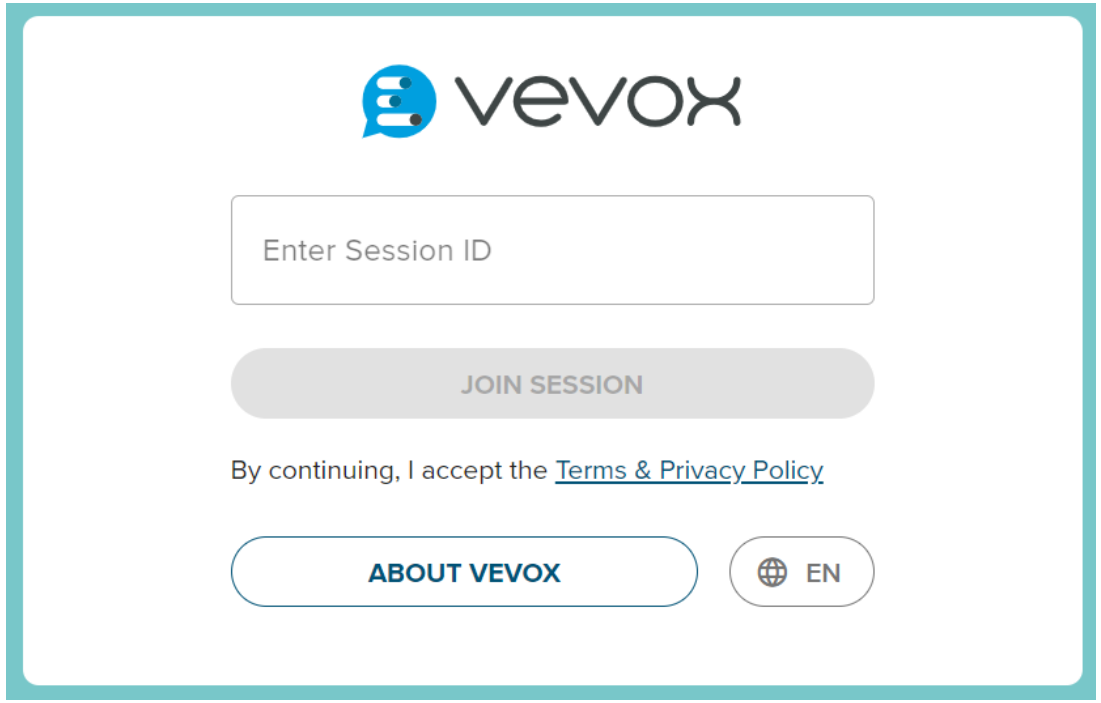
https://vevox.app

- Enter session ID

136-982-552

Enter your first and second name

- OR Enter your student ID
- OR Enter both/all



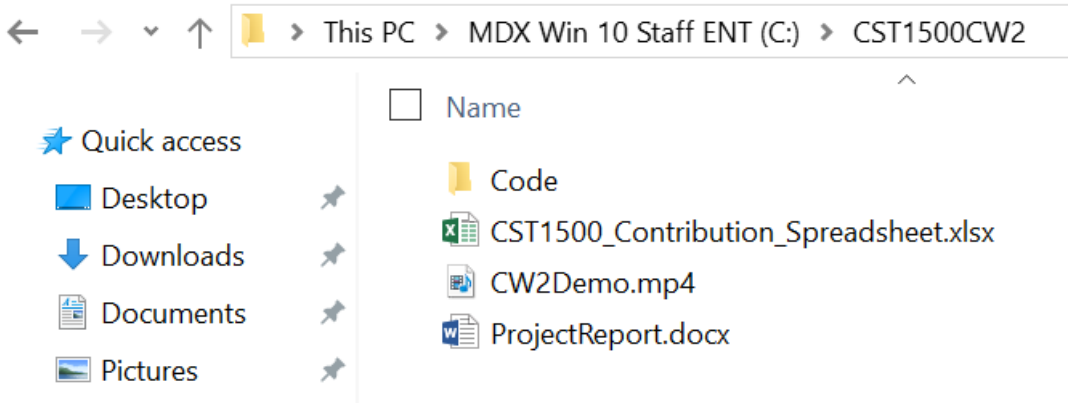
Coursework 2 – Python Groupwork

- IMPORTANT
- One member of the group should make the submission – the group should decide who makes the submission:
  - Make sure the work is completed on time.
  - Give the nominated group member enough time to collate everything and submit.

- DEADLINE
- December Friday 13th December 18:00 (2 hour window -> 20:00)

Coursework 2 – Python Groupwork

- What to submit?
  - ALL of the code – 1, 2, or 3 CPU Scheduling Algorithms.
  - ONE Word or PDF report document.
  - ONE Completed Contribution Spreadsheet.
  - ONE video demonstration.



Coursework 2 – Python Groupwork

- Code:
  - FCFS.py
  - SJF.py
  - RR.py
- Report:
  - Word .docx
  - PDF
- Contribution Spreadsheet:
  - MUST be filled in correctly i.e. no errors in the ‘Contribution Check’ column
- Video demonstration:
  - ONE .mp4 format video

Coursework 2 – Python Groupwork

- Code:
  - FCFS.py
  - SJF.py
  - RR.py
- Should have useful, descriptive comments throughout – especially for complex classes, methods/functions and anything you want to highlight.
- Should be correctly formatted using sensible variable/class/method/function names using the Python naming conventions (https://peps.python.org/pep-0008/)

Coursework 2 – Python Groupwork

- Report/Documentation:
  - Sections as stated in the CW Specification – can have subsections – MUST HAVE the sections specified in the CW Spec ->.
  - Good presentation -> Word Demo.

- Contribution Spreadsheet:
  - Complete all required fields ->.
  - You will receive your marks in a Contribution Spreadsheet PDF via email.

Coursework 2 – Python Groupwork

- Video demonstration:
  - 100MB upload limit on submissions.
  - .mkv can be very large and could prevent you from submitting.
  - IF your video is larger than 80MB – upload it to a video platform such as YouTube and clearly include the link in your Report Document.
  - Remember 10 Minutes is the maximum length – IF you go over the 10 minutes, we will not view the additional material.
  - 3:20 should be enough for each person to adequately explain their code, run it and explain the output.
  - Content will depend on complexity of code.
  - What you say will reflect your understanding of the code.
  - Try not to just read the comments that are in the code.

Coursework 2 Questions

FCFS CPU Scheduling Algorithm

- First Come First Served also known as FIFO (First In First Out).
- Simply queues processes according to the order they arrive in the ready queue.
- The process that comes first will be executed first.
- The next process starts only after the previous process is fully executed.

FCFS Advantages and Disadvantages

- Advantages:
  - Simplest scheduling algorithm for process execution.
  - Implementation of the algorithm is very easy.
  - Processes are executed according to the order of arrival.
- Disadvantages:
  - FCFS is a <u>non-preemptive</u> type of algorithm -> once the resources (CPU cycles) are allocated to a process, the process holds the CPU until it is terminated (complete) or reaches a waiting state.
  - Waiting time is generally high due to the sequential ordering of processes.
  - Easy implementation = poor performance.
  - A big processes will make small processes wait (which algorithm could be better for this situation?).

FCFS Attributes

- Arrival Time – The time at which the process arrives in the ready queue.
- Completion Time – the time at which the process completes its execution.
- Burst Time – duration of the process – how long the process lasts.
- Turn Around Time - Time Difference between completion time and arrival time. Turn Around Time = (Completion Time – Arrival Time).
- Waiting time - Time Difference between turnaround time and burst time. Waiting Time = (Turn Around Time – Burst Time).

FCFS Table Example

Process No.	Burst time	Waiting time	Turn around time
1	21	0	21
2	3	21	24
3	6	24	30
4	2	30	32

Average Waiting Time = total waiting time / number of processes
Average Turn Around Time = total turn around time / number of processes

*You can add other statistics/attributes if you want to…

FCFS Python Code

- Let’s take a look at some FCFS code examples in relation to the marking criteria ->:

Questions